

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

CS 458 - Data Mining (DM)

Fall 2025

**Lecture 6b: Improving Apriori,
Evaluation of Rules**

The Apriori Algorithm (Pseudo-Code)

C_k : Candidate itemset of size k

L_k : frequent itemset of size k

$L_1 = \{\text{frequent items}\};$

for ($k = 1; L_k \neq \emptyset; k++$) **do begin**

C_{k+1} = candidates generated from L_k ;

for each transaction t in database **do**

increment the count of all candidates in C_{k+1} that are contained in
 t

L_{k+1} = candidates in C_{k+1} with min_support

end

return $\cup_k L_k$;

Exercise

Table 6.1 Transactional Data for an *AllElectronics* Branch

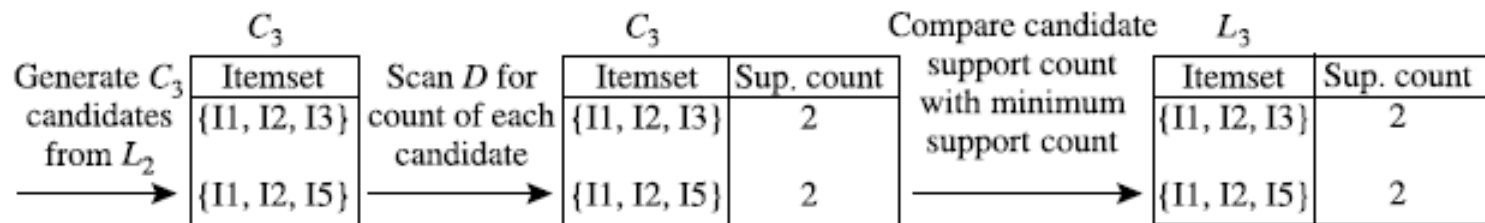
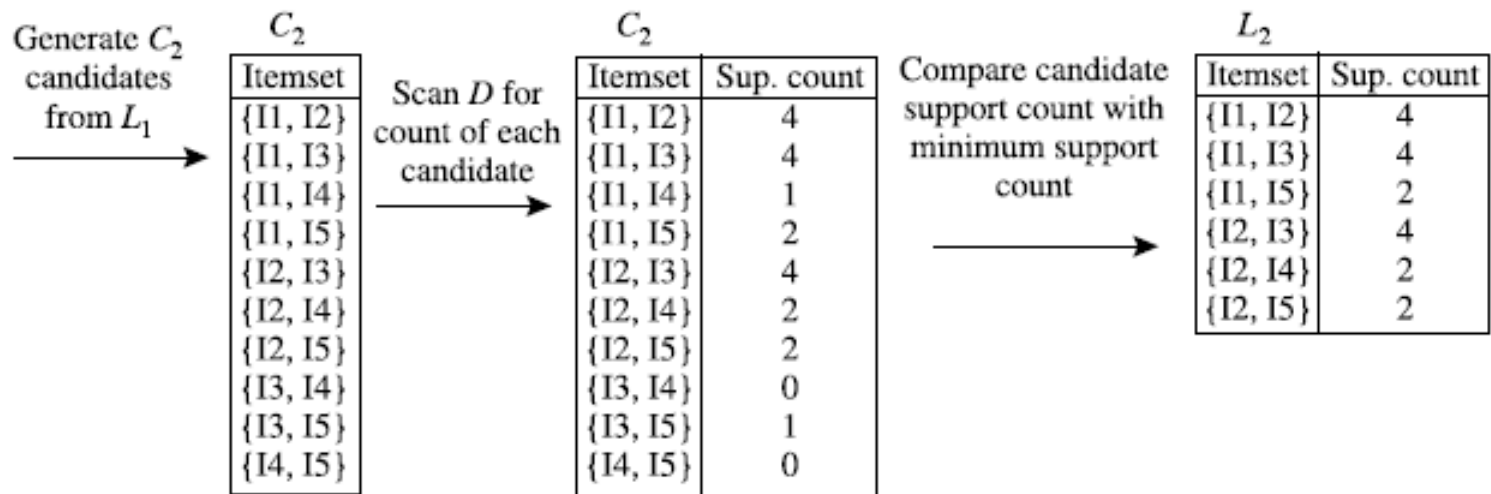
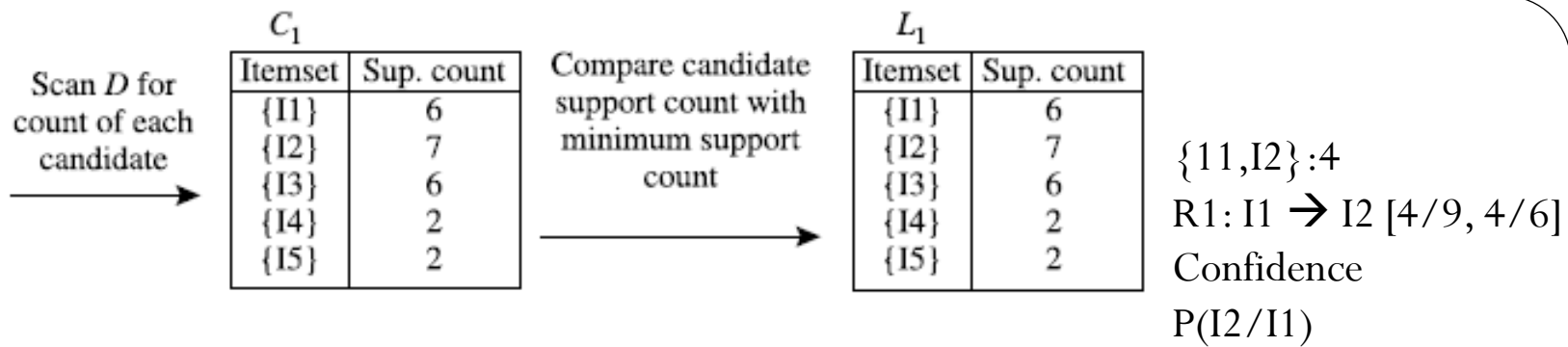
<i>TID</i>	<i>List of item_IDs</i>
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

9 transactions

5 items

Let minimum support = 2

illustrate the Apriori algorithm for finding frequent itemsets in *D*.



Generation of the candidate itemsets and frequent itemsets, where the minimum support count is 2.

Generating Association Rules from Frequent Itemsets

- For each frequent itemset l , generate all nonempty subsets of l .
- For every nonempty subset s of l , output the rule
 - $s \rightarrow l - s$

$\{I1, I2, I5\}$

$I1 \rightarrow I2, I5$ [support $2/9$, confidence $2/6 = 33\%$]

$I2 \rightarrow I1, I5$ [$2/9, 2/7$]

$I5 \rightarrow I1, I2$ [$2/9, 2/2 = 100\%$]

$I1, I2 \rightarrow I5$ [$2/9, 2/4$]

$I1, I5 \rightarrow I2$ [$2/9, 2/2$]

$I2, I5 \rightarrow I1$ [$2/9, 2/2$]

Generating association rules. Let's try an example based on the transactional data for *AllElectronics* shown before in Table 6.1. The data contain frequent itemset $X = \{I1, I2, I5\}$. What are the association rules that can be generated from X ? The nonempty subsets of X are $\{I1, I2\}$, $\{I1, I5\}$, $\{I2, I5\}$, $\{I1\}$, $\{I2\}$, and $\{I5\}$. The resulting association rules are as shown below, each listed with its confidence:

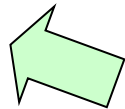
$\{I1, I2\} \Rightarrow I5,$	$confidence = 2/4 = 50\%$
$\{I1, I5\} \Rightarrow I2,$	$confidence = 2/2 = 100\%$
$\{I2, I5\} \Rightarrow I1,$	$confidence = 2/2 = 100\%$
$I1 \Rightarrow \{I2, I5\},$	$confidence = 2/6 = 33\%$
$I2 \Rightarrow \{I1, I5\},$	$confidence = 2/7 = 29\%$
$I5 \Rightarrow \{I1, I2\},$	$confidence = 2/2 = 100\%$

If the minimum confidence threshold is, say, 70%, then only the second, third, and last rules are output, because these are the only ones generated that are strong. Note that, unlike conventional classification rules, association rules can contain more than one conjunct in the right side of the rule. ■

Hands-on Association Rule Mining

- Weka
 - <https://www.cs.waikato.ac.nz/ml/weka/>
- ARFF file format
 - <https://www.cs.waikato.ac.nz/ml/weka/arff.html>
- Association Rule Mining on the transactions data set

Scalable Frequent Itemset Mining Methods

- Apriori: A Candidate Generation-and-Test Approach
- Improving the Efficiency of Apriori 
- FPGrowth: A Frequent Pattern-Growth Approach
- ECLAT: Frequent Pattern Mining with Vertical Data Format

Further Improvement of the Apriori Method

- Major computational challenges
 - Multiple scans of transaction database
 - Huge number of candidates
 - Tedious workload of support counting for candidates
- Improving Apriori: general ideas
 - Reduce passes of transaction database scans
 - Shrink number of candidates
 - Facilitate support counting of candidates

Partition: Scan Database Only Twice

- Any itemset that is potentially frequent in DB must be frequent in at least one of the partitions of DB
 - Scan 1: partition database and find local frequent patterns
 - Scan 2: consolidate global frequent patterns

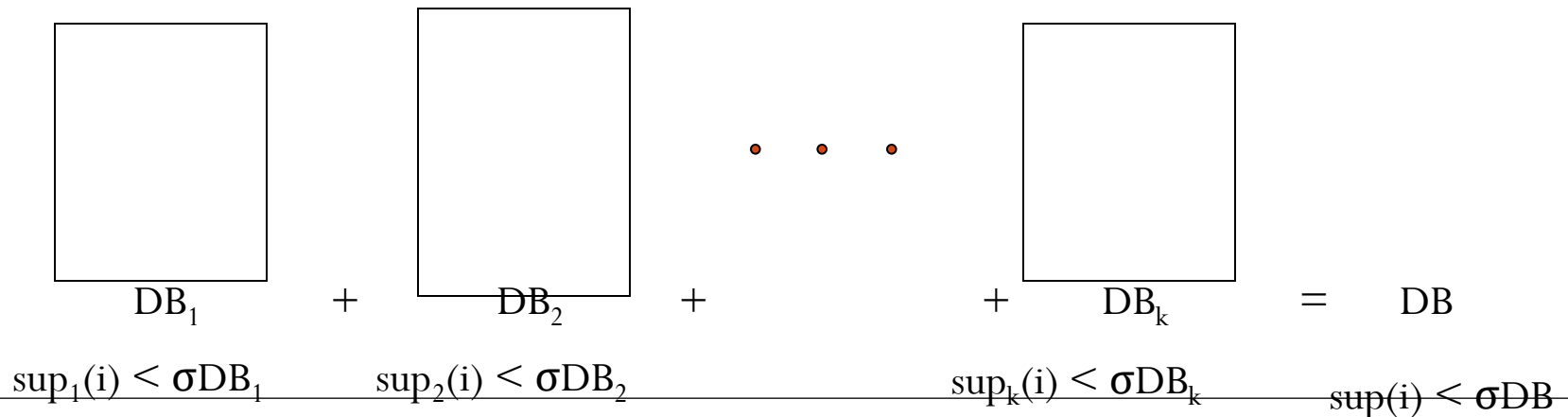
G: 100 Transactions dataset: minsup:4

P1: 50 transactions: minsup:2

P2: 50 transactions: minsup:2

Frequent itemsets from P1

Frequent itemsets from P2



G: 100 Transactions dataset: minsup:4
 $\{a,b\}:3$ (infrequent)

P1: 50 transactions: minsup:2
 $\{a,b\}:1$ will be infrequent

P2: 50 transactions: minsup:2
 $\{a,b\}:2$ frequent

Frequent itemsets from P1

Frequent itemsets from P2

Can we miss any global frequent pattern? NO

Can we get any extra frequent patterns? YES

Closed Patterns and Max-Patterns

- A long pattern contains a combinatorial number of sub-patterns, e.g., $\{a_1, \dots, a_{100}\}$ contains $\binom{100}{1} + \binom{100}{2} + \dots + \binom{100}{100} = 2^{100} - 1 = 1.27 \times 10^{30}$ sub-patterns!
- Solution: Mine *closed patterns* and *max-patterns* instead
- An itemset X is **closed** if X is frequent and there exists no super-pattern Y $\supsetneq$ X, with the same support as X
- An itemset X is a **max-pattern** if X is frequent and there exists no frequent super-pattern Y $\supsetneq$ X
- Closed pattern is a lossless compression of freq. patterns
 - Reducing the # of patterns and rules

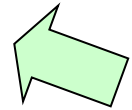
Closed and Max Pattern Example

IDs	Items
T1	{A, B}
T2	{A, B}
T3	{A}
T4	{B}
T5	{A, B}

- Closed Patterns ?
- Maximal Patterns ?

Scalable Frequent Itemset Mining Methods

- Apriori: A Candidate Generation-and-Test Approach
- Improving the Efficiency of Apriori
- ECLAT: Frequent Pattern Mining with Vertical Data Format



ECLAT: Mining by Exploring Vertical Data Format

- **ECLAT** (Equivalence Class Transformation)
- Vertical format: $t(AB) = \{T_{11}, T_{25}, \dots\}$
 - tid-list: list of trans.-ids containing an itemset
- Deriving frequent patterns based on vertical intersections
 - $t(X) = t(Y)$: X and Y always happen together
 - $t(X) \subset t(Y)$: transaction having X always has Y
- Eclat (Zaki et al. @KDD'97)
- Mining Closed patterns using vertical format: CHARM (Zaki & Hsiao @SDM'02)

Vertical Data Format

Table 6.1 Transactional Data for an *AllElectronics* Branch

<i>TID</i>	<i>List of item IDs</i>
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

Table 6.3 The Vertical Data Format of the Transaction Data Set *D* of Table 6.1

<i>itemset</i>	<i>TID_set</i>
I1	{T100, T400, T500, T700, T800, T900}
I2	{T100, T200, T300, T400, T600, T800, T900}
I3	{T300, T500, T600, T700, T800, T900}
I4	{T200, T400}
I5	{T100, T800}

T100 I1, I2, I5

T200 I2, I4

T300

T400

Support (I1) = 6

Support(I1,I2) = 4

I1 intersection I2

= {T100, T400, T800, T900}

Table 6.4 2-Itemsets in Vertical Data Format

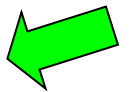
<i>itemset</i>	<i>TID_set</i>
{I1, I2}	{T100, T400, T800, T900}
{I1, I3}	{T500, T700, T800, T900}
{I1, I4}	{T400}
{I1, I5}	{T100, T800}
{I2, I3}	{T300, T600, T800, T900}
{I2, I4}	{T200, T400}
{I2, I5}	{T100, T800}
{I3, I5}	{T800}

Table 6.5 3-Itemsets in Vertical Data Format

<i>itemset</i>	<i>TID_set</i>
{I1, I2, I3}	{T800, T900}
{I1, I2, I5}	{T100, T800}

Chapter 6 Mining Frequent Patterns, Association and Correlations: Basic Concepts and Methods

- Basic Concepts
- Frequent Itemset Mining Methods
- Which Patterns Are Interesting?—Pattern Evaluation Methods
- Summary



Misleading strong rule

- $N = 10000$
- Games = 6000
- Videos = 7500
- Games and Videos = 4000
- Min_Sup = 30%, Min_Conf = 60%
- Games $\rightarrow$ Videos
 - [support = 40%, Confidence = $P(V / G) = P(V \text{ and } G) / P(G) = 66.7\% = 4000 / 6000$]
- $P(\text{Videos}) = .75 = 7500 / 10000$
- $P(\text{Videos} / \text{Games}) = 0.67$
- Infact computer games and videos are negatively associated
- Lift = ?

	Game	~Game	Sum (row)
Video	4000	3500	7500
~Video	2000	500	2500
Sum(col.)	6000	4000	10000

Interestingness Measure: Correlations (Lift)

- *Buy Games* $\Rightarrow$ *BuyVideos* [40%, 66.7%] is misleading
 - The overall % of video is 75% > 66.7%.
- *Buy Games* $\Rightarrow$ *not videos* [20%, 33.3%] is more accurate, although with lower support and confidence
- Measure of dependent/correlated events: lift

$$lift = \frac{P(A \cup B)}{P(A)P(B)}$$

$$lift(G, V) = \frac{4000/10000}{6000/10000 * 7500/10000} = 0.89$$

$$lift(G, \neg V) = \frac{2000/10000}{6000/10000 * 2500/10000} = 1.33$$

	Game	~Game	Sum (row)
Video	4000	3500	7500
~Video	2000	500	2500
Sum(col.)	6000	4000	10000

$P(A \text{ and } B) = P(A) * P(B)$, when A and B are independent
Lift = $P(A)P(B) / P(A)P(B) = 1$

Basic Concepts: Association Rules

Tid	Items bought
10	cola, Nuts, Diaper
20	cola, Coffee, Diaper
30	cola, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk

- $cola \rightarrow Diaper$ (60%, 100%)
- $Lift = .6 / (.6)(0.8) = 1.25 > 1$ means positive correlation
- $P(Diaper/Cola) = 100\%$
- $P(Diaper) = 4/5 = 80\%$
- $Diaper \rightarrow cola$ (60%, 75%)

Table 6.6 2×2 Contingency Table Summarizing the Transactions with Respect to Game and Video Purchases

	<i>game</i>	$\overline{\text{game}}$	Σ_{row}
<i>video</i>	4000	3500	7500
$\overline{\text{video}}$	2000	500	2500
Σ_{col}	6000	4000	10,000

Table 6.7 Table 6.6 Contingency Table, Now with the Expected Values

	<i>game</i>	$\overline{\text{game}}$	Σ_{row}
<i>video</i>	4000 (4500)	3500 (3000)	7500
$\overline{\text{video}}$	2000 (1500)	500 (1000)	2500
Σ_{col}	6000	4000	10,000

$$6000 * 7500 / 10000 = 4500$$

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} = \frac{(4000 - 4500)^2}{4500} + \frac{(3500 - 3000)^2}{3000} \\ + \frac{(2000 - 1500)^2}{1500} + \frac{(500 - 1000)^2}{1000} = 555.6.$$

Because the χ^2 value is greater than 1, and the observed value of the slot (*game, video*) = 4000, which is less than the expected value of 4500, *buying game* and *buying video* are *negatively correlated*. This is consistent with the conclusion derived from the analysis of the *lift* measure in Example 6.8. ■